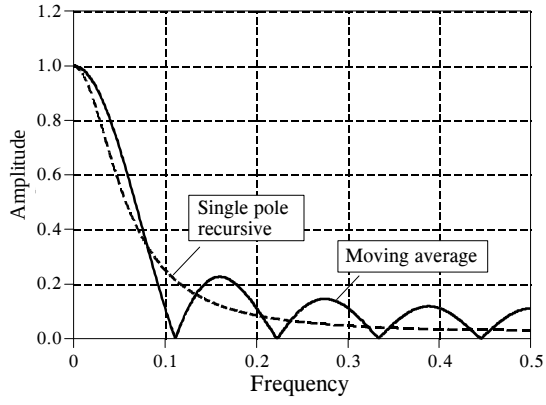

FIGURE 21-6
Moving average and single pole frequency responses. Both of these filters have a poor frequency response, as you should expect for time domain filters.

two cases where one filter has a slight edge over the other. These are based on the trade-off between *development* time and *execution* time. In the first instance, you want to reduce development time and are willing to accept a slower filter. For example, you might have a one time need to filter a few thousand points. Since the entire program runs in only a few seconds, it is pointless to spend time optimizing the algorithm. Floating point will almost certainly be used. The choice is to use the moving average filter carried out by convolution, or a single pole recursive filter. The winner here is the recursive filter. It will be slightly easier to program and modify, and will execute much faster.

The second case is just the opposite; your filter must operate as fast as possible and you are willing to spend the extra development time to get it. For instance, this filter might be a part of a commercial product, with the potential to be run *millions* of times. You will probably use integers for the highest possible speed. Your choice of filters will be the moving average

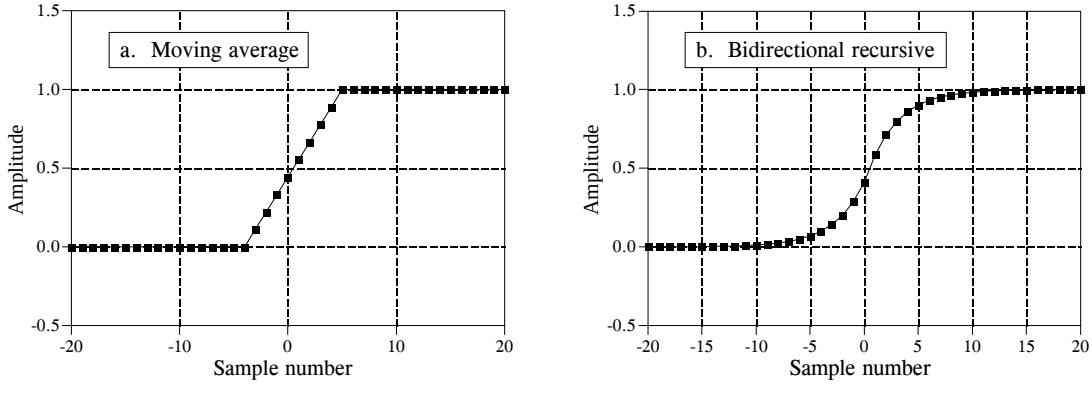

FIGURE 21-7
Step responses of the moving average and the bidirectional single pole filter. The moving average step response occurs over a smaller number of samples, while the single pole filter's step response is smoother.